Name: _____

Instructions: This lab is a diagnostic assignment. It is not graded and will not count toward your course grade. The purpose is to give both you and your instructor a sense of your current comfort level with Desmos and with using technology in calculus. Please complete the assignment as honestly as possible without outside assistance, so the results reflect your actual starting point. Do not use calculators, Wolfram Alpha, ChatGPT, or other technology tools to complete this assignment. **Use only your brain and Desmos.** Show your work neatly and clearly on this packet. If a problem asks for a graph or plot, generate it in Desmos and include a copy with your submission. Submit your completed work by uploading it to D2L by the due date listed below. This is for diagnostic purposes only. Your work will help guide instruction and will not be graded for correctness.

Diagnostic Desmos Lab

Due: Friday, September 5

1. Plotting a Function and its Inverse: Suppose you deposit \$15 into a savings account that yields 3% annual interest and it is compounded quarterly (i.e. every three months). Then the total amount, call it $A(t)$, in your bank account t years after the initial deposit is given by the following exponential function

$$A(t) = 15\left(1 + \frac{0.03}{4}\right)^{4t}$$

   (a) Go to www.desmos.com and create a Desmos account, and start graphing. Define this function in Desmos as $A(t)$ and plot it on a window for $x \in [-5, 50]$ and $y \in [-2, 80]$. Submit a screenshot of this graph on D2L.

   (b) How much money will be in your account after 20 years? Use Desmos to calculate this value.

   (c) The annual percentage yield (APY) for this savings account is 3%. In the function for $A(t)$, change the value 0.03 to p. Desmos will ask you if you want to make a slider for p, click yes. Change the bounds on p to go from 0.03 to 0.15 with a step size of 0.01. Slide p to go from 0.03 to 0.15. Describe what happens to the graph of $A(t)$. Explain why this behavior makes sense.

   (d) Let $p = 0.15$, which is an APY of 15%. How much money is in your account after 20 years?

(e) The number of times interest is compounded in this savings account is 4 times per year. In the function for $A(t)$, change every instance of 4 to n, and add a slider for n that ranges from 1 to 365 with a step size of 1. What happens to the function $A(t)$ as n increases? Is it better if your money is compounded more times per year? Explain.

(f) Letting $p = 0.03$, report the amount of money in the account after 20 years when $n = 1$ and when $n = 365$. Does this support your answer to part (e)?

(g) Use the space below to find the inverse of this function, solving for t in terms of A. Note: your function for t should depend on A, and your expression should also involve the parameters p and n. For any given value of A, describe what the output of this function in one sentence.

(h) Graph the function $t(A)$ in Desmos. Adjust the window to be a square that nicely shows both $A(t)$ and $t(A)$. Describe the relationship between these two functions. Submit a screenshot of both graphs on D2L.

(i) Let $p = .03$ and $n = 4$. Using Desmos and your function for $t(A)$, determine how long it will take for your initial \$15 to double. How long will it take to have \$100 in your account?

(j) Let $n = 4$. In the space below, find the APY (p) needed in order for your \$15 to double in 2 years. Share the link to your Desmos page on D2L that validates your answer for p.